

APPLIED DATA SCIENCE CAPSTONE

SpaceX Falcon 9 First-Stage Landing Prediction

Can we predict whether the Falcon 9 first stage will land successfully — and use that to estimate the true cost of a launch?

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github.com/Patrick-Borj-Tuano/IBM-Data-Science

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SpaceX Cuts Launch Costs by Reusing Rocket Boosters

SpaceX advertises Falcon 9 launches at **\$62 million**, versus \$165 million+ charged by other providers — largely because SpaceX **reuses the first stage** instead of discarding it. If a competing launch provider ("SpaceY") can predict, before launch, whether that first stage will land successfully, it can estimate the true cost of a launch and compete on price.

90

Launches analyzed

Falcon 9 missions collected via API +
Wikipedia scraping

66.7%

Historical success rate

of first-stage landing attempts (Class
= 1)

88.9%

Best training accuracy

achieved by a tuned Decision Tree
classifier

83.3%

Best held-out test accuracy

shared by Logistic Regression, SVM,
and KNN

Bottom line: launch outcome is learnable from pre-flight data (orbit, payload mass, launch site, flight history) — giving SpaceY a data-driven path to competitive pricing.

The Business Problem & Project Objectives

Since 1957, spacefaring nations and companies have competed to launch payloads into orbit — at an average cost of \$165M per launch. SpaceX broke that pricing model by recovering and reusing the Falcon 9 first stage. Successful recovery hinges on a controlled landing, either on a drone ship or a ground pad.

As a data scientist for a new entrant, SpaceY, the objective is to determine — from publicly available launch data — which variables (payload mass, orbit type, launch site, flight number, booster version) predict a successful landing, and to build a classifier that estimates that probability before launch.

Key Questions This Project Answers

- How does payload mass affect landing success?
- Do success rates differ by launch site?
- Do success rates differ by orbit type?
- Has the success rate improved over time (2010–2020)?
- What single algorithm best classifies landing outcome, and how well?

Data Collection: API + Web Scraping

Two complementary sources were combined into one clean launch-level dataset.

SpaceX REST API

Requested launch records in JSON via the public SpaceX API, then parsed rocket, payload, launch site, core, and landing-outcome fields into a pandas DataFrame. Rows with missing PayloadMass were imputed with the column mean.

GitHub: [notebook](#) 

Wikipedia Web Scraping

Sent an HTTP GET request to the “List of Falcon 9 and Falcon Heavy launches” Wikipedia page, then used BeautifulSoup to parse the HTML launch table into structured rows, extracting flight number, date, booster version, and outcome.

GitHub: [notebook](#) 

Result: a merged flight-by-flight dataset spanning 90 Falcon 9 launches (2010–2020), each row describing one launch attempt and its landing outcome.

Data Wrangling

With pandas and NumPy, the raw merged dataset was cleaned and a binary training label was engineered from the free-text landing outcome.

- 1 Loaded the collected dataset and checked column data types
- 2 Quantified missing values per column (LandingPad: 40.6% missing — expected, since many attempts had no landing pad)
- 3 Split attributes into numerical vs. categorical
- 4 Counted launches per site — CCAFS SLC-40 (55), KSC LC-39A (22), VAFB SLC-4E (13)
- 5 Counted occurrences of each orbit type — GTO (27), ISS (21), VLEO (14), and 8 others
- 6 Engineered a binary Class label (1 = successful landing) from the 8 raw Outcome strings, e.g. 'True ASDS' → 1, 'False ASDS' → 0
- 7 Computed the overall first-stage landing success rate: 66.7%

GitHub: [Data Wrangling lab.ipynb](#) 

EDA & Interactive Visual Analytics

EDA with Visualization

seaborn / matplotlib scatter, bar, and line plots to surface relationships between flight number, payload mass, orbit, launch site, and landing outcome.

EDA with SQL

Loaded the dataset into a Db2 table (SPACEXTBL) and ran 10 SQL queries — filtering, aggregation, subqueries — to answer specific business questions.

Interactive Map (Folium)

Plotted all 4 launch sites as map markers, color-coded launch outcomes per site, and drew polylines measuring each site's proximity to coastlines, highways, railways, and cities.

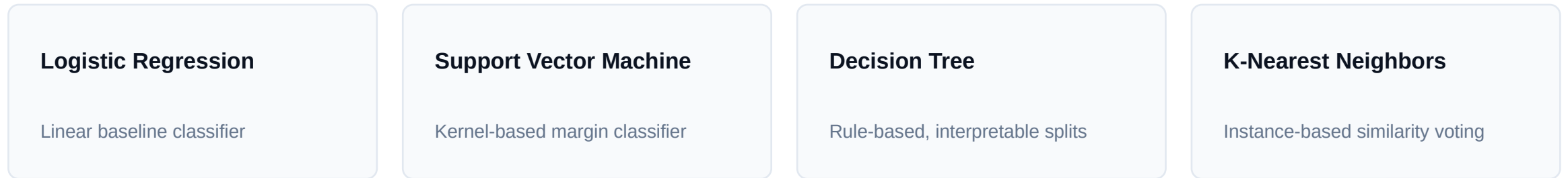
Dashboard (Plotly Dash)

Built an interactive dashboard with a launch-site dropdown, a success-rate pie chart, and a payload-vs-outcome scatter chart driven by a payload-range slider.

Predictive Analysis (Classification)



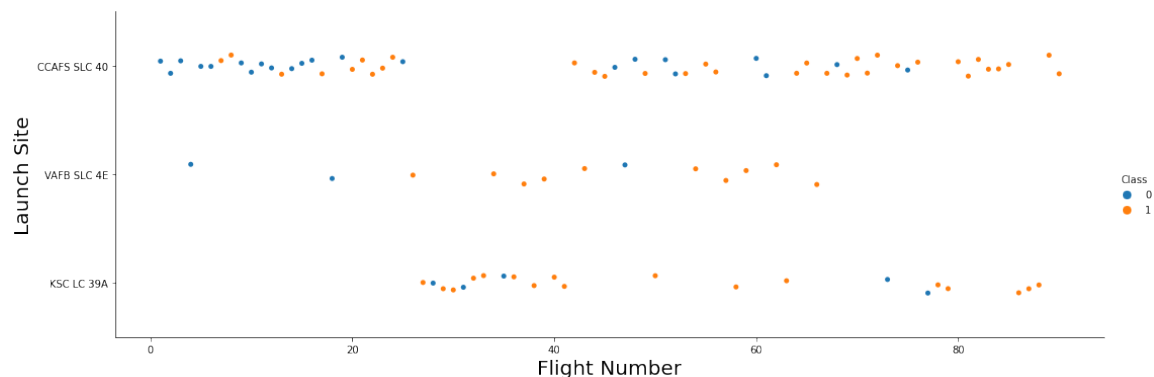
Four classification algorithms were tuned and compared:



Hyperparameters were tuned with GridSearchCV (10-fold CV); models were evaluated on the held-out test set using accuracy, F1 score, and Jaccard score, with confusion matrices for error analysis.

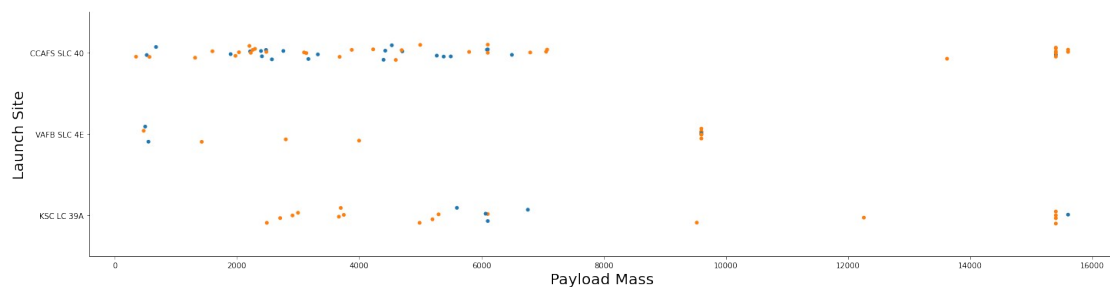
GitHub: [Machine Learning Prediction lab.ipynb](#) 

Visualization: Site & Payload Effects



Flight Number vs. Launch Site (colored by outcome)

Later flights succeed far more often, across every site — evidence of a learning curve as SpaceX refined its recovery process.



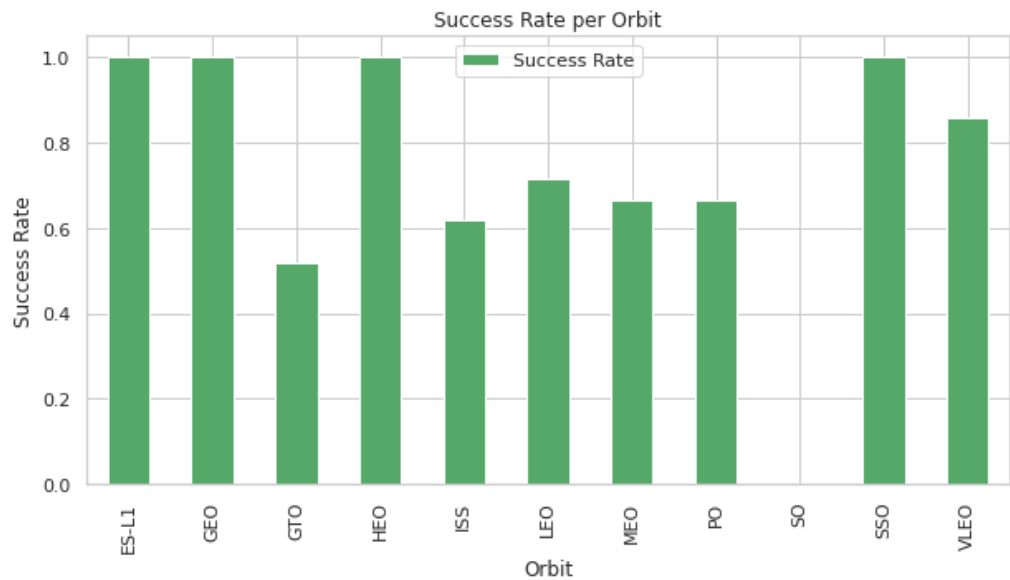
Payload Mass vs. Launch Site

What the charts show

- CCAFS SLC-40: 55 attempts, ~60% success — highest volume, moderate rate
- VAFB SLC-4E: 13 attempts, ~77% success — highest rate, lowest volume
- KSC LC-39A: 22 attempts, ~77% success
- No strong linear relationship between raw payload mass and landing success — outcome depends more on orbit and booster generation than mass alone

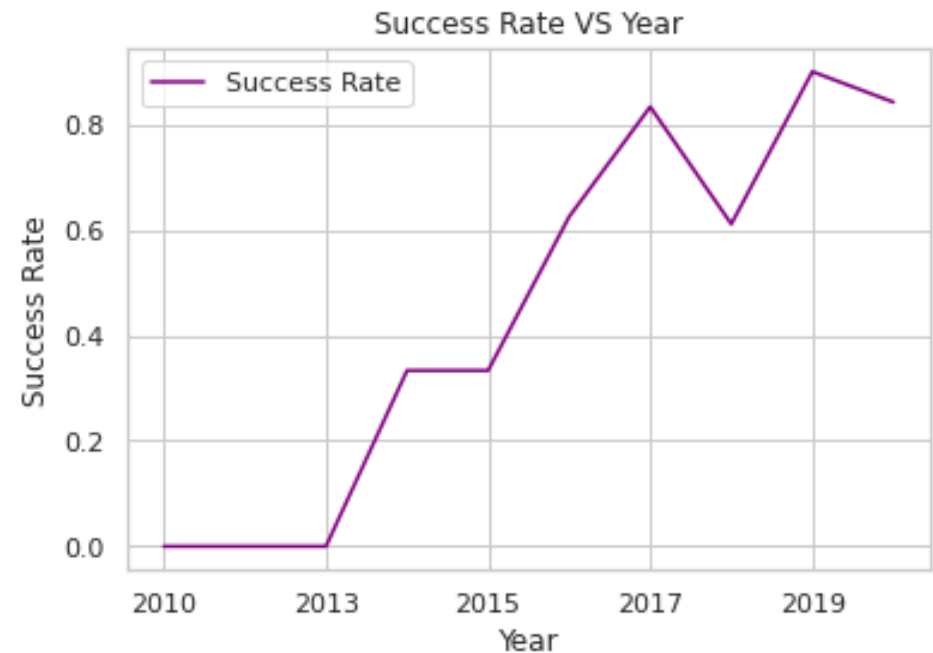
GitHub: [EDA with Visualization lab.ipynb](#)

Visualization: Orbit Type & Yearly Trend



Success Rate by Orbit Type

ES-L1, GEO, HEO, and SSO orbits show the strongest first-stage recovery; GTO (geostationary transfer) is the hardest to recover from — these missions demand the most energy, leaving the least fuel margin for a controlled landing.



Launch Success Rate by Year

Success rate climbs steadily from 2013 onward, reaching near 100% by 2019–2020 — the clearest single signal that recovery technology matured over time.

SQL Query Insights (IBM Db2)

The cleaned dataset was loaded into a Db2 table (SPACEXTBL) and queried directly to answer specific business questions.

Query	Result
Total NASA (CRS) payload mass	45,596 kg
Avg. payload mass, booster F9 v1.1	2,928 kg
First successful ground-pad landing	2015-12-22
Mission outcomes	99 Success / 1 Failure (in flight) / 1 Success (payload status unclear)
Max-payload booster versions	F9 B5 B1048.4/.5, B1049.4/.5/.7, B1051.3/.4/.6, B1056.4, B1058.3, B1060.2/.3
Top landing outcome, 2010–2017	No attempt (10), Failure – drone ship (5), Success – drone ship (5)

GitHub: [EDA with SQL lab.ipynb](#) 

Interactive Map Results (Folium)

4 Falcon 9 Launch Sites

	CCAFS LC-40	28.5623, -80.5774
	CCAFS SLC-40	28.5632, -80.5768
	KSC LC-39A	28.5733, -80.6469
	VAFB SLC-4E	34.6328, -120.6107

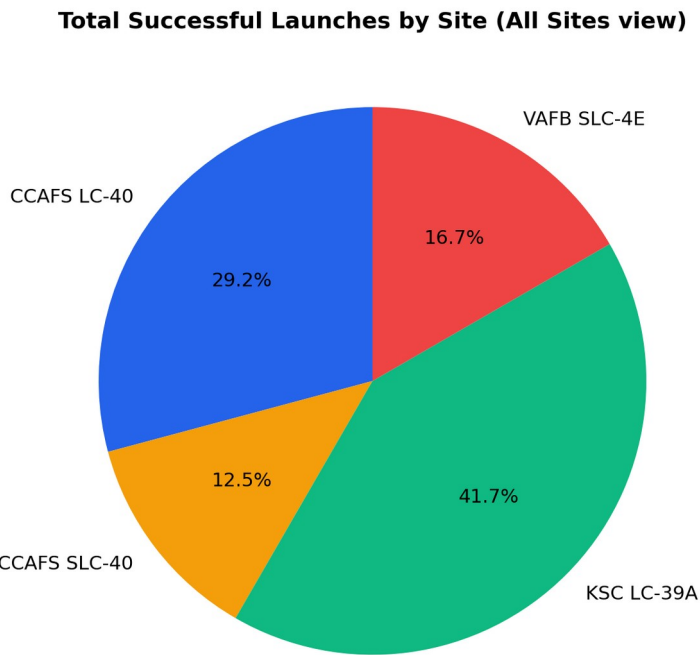
All 4 launch pads sit within a few kilometers of a coastline (closest ≈ 1.32 km from CCAFS LC-40), close to highways and railways for logistics, yet a deliberate distance from the nearest city (Orlando, ~ 50 km) — confirming a safety buffer while keeping transport short.

GitHub: [Folium lab.ipynb](#) 

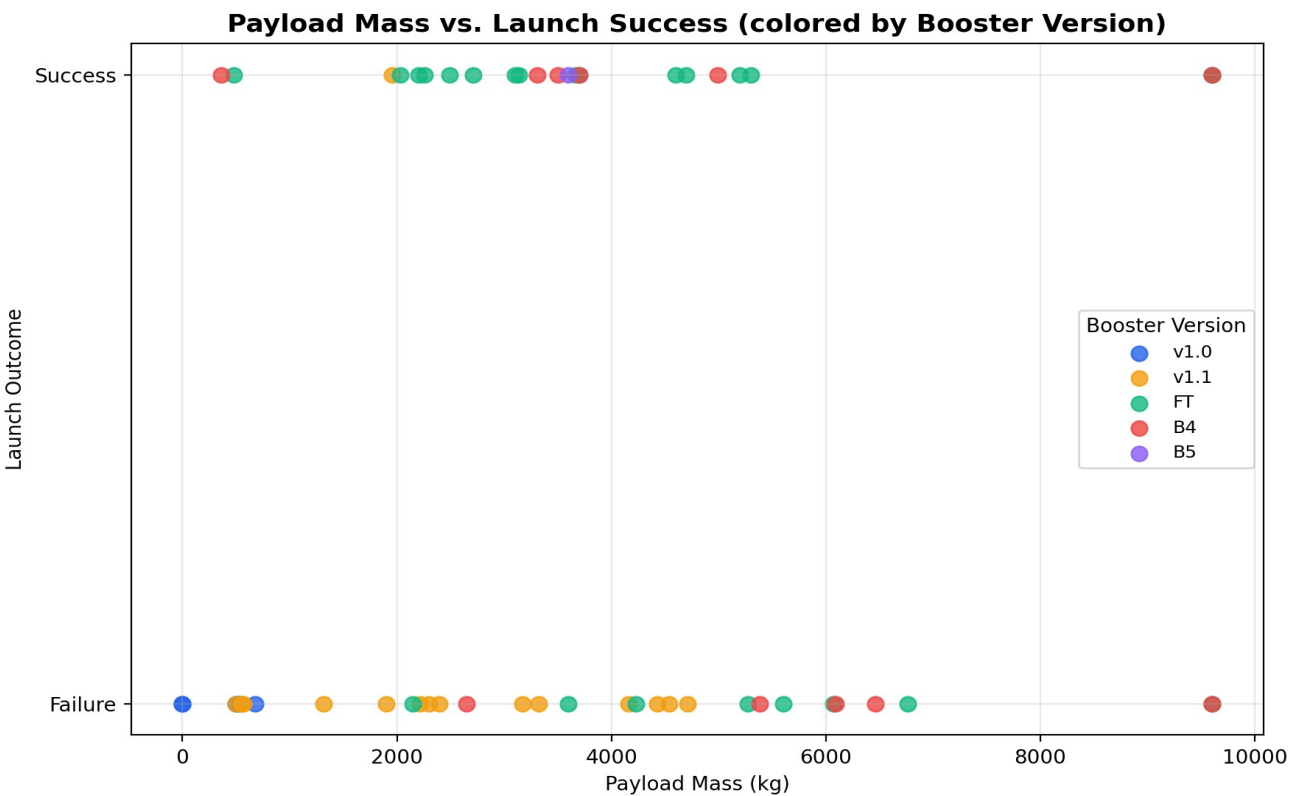
Built with Folium

- Circle + marker for each launch site, popped up with site name
- MarkerCluster color-codes individual launch outcomes (green = success, red = failure) at each site
- MousePosition plugin for live lat/long readout while exploring
- PolyLine distance measurements to the nearest coastline, highway, railway, and city

Interactive Dashboard Results (Plotly Dash)



Success-share pie chart (site dropdown drives this view)



Payload vs. outcome scatter (payload-range slider filters this view)

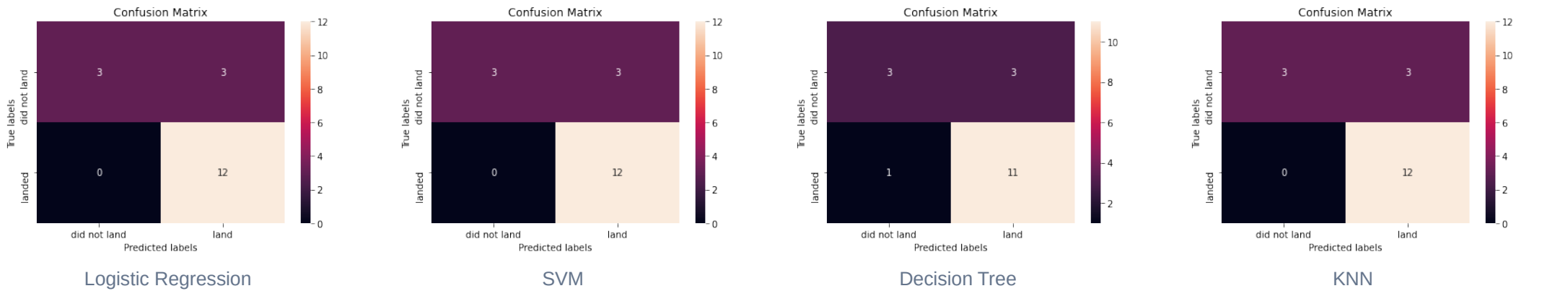
KSC LC-39A leads all sites with a 77% success share; the scatter confirms outcome is driven more by booster generation (color) than by payload mass alone — later booster versions (v1.1, FT, B4, B5) succeed across a wide payload range.

Classification Results

Model	Best Params (Grid Search)	Train Acc.	Test Acc.
Logistic Regression	C=0.01, penalty=l2, solver=lbfgs	84.6%	83.3%
SVM	kernel=sigmoid, C=1.0, γ=0.032	84.8%	83.3%
Decision Tree	criterion=entropy, max_depth=6	88.9%	77.8%
KNN	n_neighbors=10, p=1	84.8%	83.3%

Logistic Regression, SVM, and KNN tie for best generalization at 83.3% test accuracy; the Decision Tree overfits (highest train accuracy, lowest test accuracy).

Confusion Matrices (test set)



CONCLUSION

What SpaceY Learns From This Data

- 1 First-stage landing outcome is predictable from pre-flight variables — flight number, orbit, launch site, and payload mass — with ~83% held-out accuracy from simple, interpretable models.
- 2 Landing success has a strong time trend: from under 40% (2013) to near 100% (2019–2020), reflecting SpaceX's iterative engineering improvements captured naturally in the flight-number feature.
- 3 Orbit type matters more than raw payload mass: ES-L1/GEO/HEO/SSO missions land reliably; GTO missions are the highest-risk category and the biggest lever for improving recovery rate.
- 4 Model choice should favor generalization over training fit: the Decision Tree memorized the training set (88.9%) but generalized worse (77.8%) than the simpler Logistic Regression, SVM, and KNN models (83.3% each).
- 5 Recommendation for SpaceY: prioritize launch sites and orbit classes with the strongest historical recovery record when estimating bid costs, and revisit the model as more flights accumulate.

Innovative Insight: Reframing “Flight Number”

The most predictive feature in this dataset isn't a physics variable — it's FlightNumber, a simple counter. That's worth unpacking, because it reveals something the other features can't show on their own.

FlightNumber is a proxy for accumulated engineering knowledge

Every additional flight embeds lessons from every prior landing attempt — grid-fin redesigns, titanium fins, updated guidance software. The model isn't learning “later is better” by coincidence; it's implicitly learning a learning curve.

This reframes the business question for SpaceY

Instead of asking “will this exact launch succeed?”, SpaceY should ask “how many attempts until our recovery process matures like SpaceX's did?” — turning a one-shot prediction into a roadmap for R&D investment and pricing strategy over time.

Practical implication: cost estimates should be attempt-dependent

A SpaceY bid submitted in “year 1” of a reusability program should assume a materially lower recovery rate — and thus higher effective cost per launch — than a bid submitted after 60+ attempts, mirroring the trend seen in this data.

APPENDIX

GitHub Repository & Notebooks

Full repository: <https://github.com/Patrick-Borj-Tuano/IBM-Data-Science>

01 [Data Collection API Lab.ipynb](#)

03 [Data Wrangling lab.ipynb](#)

05 [EDA with Visualization lab.ipynb](#)

07 [spacex_dash_app.py](#)

02 [Data Collection with Web Scraping lab.ipynb](#)

04 [EDA with SQL lab.ipynb](#)

06 [Interactive Visual Analytics with Folium lab.ipynb](#)

08 [Machine Learning Prediction lab.ipynb](#)